Politics or Print? Separating Content from Shortcuts when Classifying Two Nineteenth-Century isiXhosa Newspapers

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Abstract

Does a classifier separating *Imvo Zabantsundu*, founded by J. T. Jabavu in 1884, from the missionary *Isigidimi sama-Xosa* read politics or print? On the isiXhosa Intellectual Traditions corpus (Nov 1884–1888) we train tuned classifiers along a ladder that removes English, self-references, names and spelling conventions, with non-content baselines, matching and noise injection. Raw text is separated with balanced accuracy 0.93, but spelling and transcription features alone reach 0.85. A model on the remaining shared vocabulary keeps 0.81 (0.78 on matched pairs): a source signal survives, but not one attributable to politics alone. A lexicon supports a political difference: Imvo uses about twice Isigidimi's political vocabulary overall, and more in every common year. On earlier Isigidimi years the classifier shows a temporal shift of unisolated cause.

Keywords: historical isiXhosa; shortcut learning; responsible NLP

Introduction

Isigidimi sama-Xosa (Lovedale, 1870–1888) was published under missionary control. *Imvo Zabantsundu* (King William's Town, from November 1884) was founded by John Tengo Jabavu, who had edited Isigidimi from 1881 (South African History Online, 2026) and left because the missionaries were reluctant to publish explicitly political material (Odendaal, 1984). The digitised archive of Schoots et al. (2023) makes the split testable: if the break was political, the papers should differ in *what* they discuss. A source classifier looks like natural evidence, but classifiers exploit whatever separates the labels (Gururangan et al., 2018; Geirhos et al., 2020), and historical text is full of such shortcuts: collection-specific transcription error (Hill & Hengchen, 2019; van Strien et al., 2020), mastheads, adverts and house spelling.

Research question. *When a classifier separates the two papers over the years both appeared, how much of the separation is content, and how much is identity, transcription noise and orthography?* We test whether (H1) raw accuracy is inflated by shortcuts, (H2) a signal survives their removal, and (H3) that residual is political rather than religious. Mistaking print for politics would yield false claims about historical Black writers in a low-resource language (Joshi et al., 2020).

Methods

Data. We use both papers from November 1884 (Imvo's first issue) to December 1888 (Isigidimi's last; its 1886 volume is lost). Pages are cut into 200-token windows (6,699 Imvo and 1,509 Isigidimi segments from 211 and 38 issues). Splits are 70/15/15 *by issue*, stratified by paper and year, for three seeds. Validation and test segments sharing over 30% of their word 6-grams with training text (reprinted adverts) are removed: 558 of 2,512, 277 of them from test.

Cleaning ladder. L0 is the full page; L1 keeps only the dataset's isiXhosa paragraphs (Imvo is 27% English; Schoots et al., 2023). L2 also removes self-references (titles, place of printing, staff), digits and punctuation. L3 also masks capitalised non-sentence-initial tokens (names). L4 is lower-cased, equalises orthography (⟨bh⟩→⟨b⟩, ⟨xh⟩→⟨x⟩, no elision apostrophes) and keeps the 3,485 words seen ≥ 5 times in *both* papers' training data, so one-paper spellings and OCR errors are unusable.

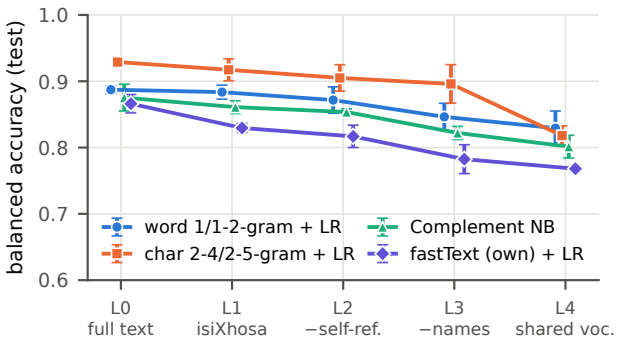


Figure 1: Test balanced accuracy (chance $= 0.5$) as shortcuts are removed; mean $\pm$ s.d. over three issue-grouped splits. The character model, which can read spelling, loses most when orthography is equalised (L3→L4).

Models. Word (unigram or uni+bigram) and character (2–4 or 2–5-gram) TF-IDF with class-balanced logistic regression, Complement Naive Bayes, and fastText subword embeddings (Bojanowski et al., 2017) trained on the training split and mean-pooled. Pretrained encoders (Alabi et al., 2022) could not be downloaded here. Hyper-parameters are tuned on validation macro-F1 (Supp. Table 4); each test set is used once. With Isigidimi at 18% of segments, we report macro-F1 and balanced accuracy (BA) with 95% CIs from a bootstrap over issues.

Isolating the signal. (i) *Feature families* on L4: the 100 most frequent shared words (cf. Mosteller & Wallace, 1964) versus the remaining shared vocabulary. Frequency separates style from topic only loosely (the top 100 include *abantu*, "people"), so a sensitivity model also drops 359 grammatical fragments and residual names. On raw text: 15 transcription-quality proxies (e.g. vowel-less tokens; not measured OCR error rates, as the archive was human-corrected) and 4 orthographic rates (⟨bh⟩, ⟨xh⟩, elision, hyphens). (ii) *Matching*: each Isigidimi test segment is paired 1:1 with its nearest Imvo segment on the 19 standardised non-content features, within a median-distance caliper (Stuart, 2010). This gives 134 pairs (mean absolute standardised difference 0.29→0.08). (iii) OCR-style *noise injection* into Isigidimi test text; (iv) a *temporal hold-out* (test on 1888). (v) *Probe*: the content model scores unseen Isigidimi issues from 1876–80 (before Jabavu) and 1881–Oct 1884 (under Jabavu). (vi) *Lexicon*: 14 political patterns (10 concepts, e.g. *palamente* parliament) and 12 religious (11 concepts, e.g. *Tixo* God), drafted before comparing the papers and audited by concordance (Supp.). Follow-ups added after seeing results are exploratory (Supp. S4).

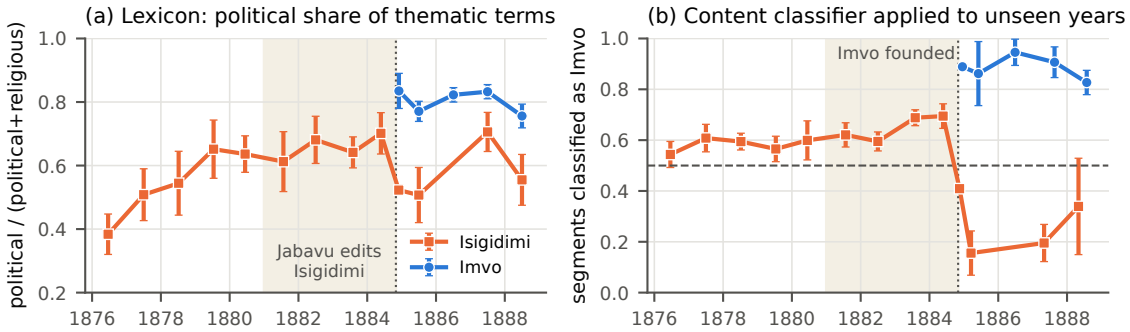


Figure 2: Isigidimi before, during (shaded) and after Jabavu's editorship, with Imvo for comparison; 1884 is split at Imvo's founding, and bars show ± 1.96 s.e. over issues. **(a)** Political share of thematic terms. **(b)** Share of segments the L4 remaining-vocabulary model (trained on Nov 1884–88) labels "Imvo"; earlier years land on the Imvo side whoever the editor was.

Table 1: Test results: macro-F1 with issue-bootstrap 95% CI (seed 13), its s.d. over 3 splits, BA, and BA_m on the 134 matched pairs (CIs in Supp.). Majority-class macro-F1 ≈ 0.45.

Representation	Macro-F1 [CI]	s.d.	BA	BA_m
Full page text, char	0.926 [0.90, 0.95]	0.010	0.92	–
isiXhosa only, char	0.916 [0.88, 0.95]	0.012	0.91	0.86
isiXhosa only, words	0.883 [0.85, 0.92]	0.009	0.88	0.85
L4 shared vocab., words	0.840 [0.79, 0.88]	0.019	0.84	0.81
L4 shared vocab., fastText	0.717 [0.68, 0.76]	0.018	0.76	–
L4 top-100 frequent words	0.718 [0.68, 0.76]	–	0.75	0.71
L4 remaining vocabulary	0.802 [0.76, 0.84]	–	0.81	0.78
minus fragments, names	0.750 [0.70, 0.80]	–	0.78	0.75
Transcription features	0.715 [0.66, 0.77]	–	0.74	0.62
Orthography features	0.759 [0.69, 0.82]	–	0.83	0.73
All non-content features	0.831 [0.78, 0.87]	–	0.85	0.75

Results

Shortcuts inflate accuracy (H1). On full pages character n-grams reach BA 0.93 ± 0.01 (Fig. 1). The strongest L1 character features are house conventions: Imvo writes ⟨bh⟩ (8.7 tokens per 1,000 vs. 0.01) and ⟨xh⟩, while Isigidimi uses elision apostrophes three times as often. Imvo also has 2.4× more vowel-less tokens, a proxy for residual transcription errors. Non-content features alone reach BA 0.85 (Table 1).

A source signal survives (H2). Cleaning generally reduces performance, yet the L4 word model keeps BA 0.83 ± 0.03 across splits and the remaining vocabulary 0.81 (0.78 without fragments and names). Matching costs this model 0.03 (to 0.78 [0.72, 0.84]) and the non-content models 0.10–0.12 (to 0.75 [0.68, 0.82]); the intervals overlap. Corrupting 5% of Isigidimi tokens flips 2–5% of segments to "Imvo". Tested on unseen 1888, the L4 model keeps BA 0.80.

Lexicon results support a political difference (H3). Top Imvo content features include *abamnyama* and *amawetu* ("black people", "our people"), *imantyi* (magistrate), *amatyala* (court cases), and *entolongweni* (in prison). Isigidimi's include *imbali* (history), *inkosi* (chief), *imfazwe* (war) and *Ntsikana*, an early Christian hymn-writer. Imvo uses 14.0 political terms per 1,000 tokens [13.2, 14.7] against Isigidimi's 7.4 [6.2, 8.7]. In each of the four years both papers appeared, Imvo's rate is higher (by 4.3–7.1; within-year permutation $p < 0.001$), also after dropping ambiguous patterns such as *tyala* ("debt" or "case"; by 3.0–6.0). Within Isigidimi (Fig. 2a), the political share rises from 0.54 [0.49, 0.58] in 1876–80 to 0.67 [0.63, 0.70] under Jabavu, then falls to 0.59 [0.54, 0.63] after he left.

The classifier is a poor clock. On unseen Isigidimi years (Fig. 2b) the content model calls 66% of Jabavu-era segments "Imvo", but also 58% of 1876–80 segments, against 26% of held-out post-1884 ones. The Jabavu-era excess over 1876–80 is small (+0.05 mean probability, CI [0.03, 0.07]).

Discussion

Print alone separates the papers (BA 0.85), so a 0.93 source classifier is no evidence of a political split (H1). After the controls a source signal near 0.8 remains (H2); the lexicon ties part of the difference to political vocabulary (H3), but genre, vocabulary and residual conventions also separate the papers. This fits the account that Jabavu left to write politics freely (Odendaal, 1984): Isigidimi's political share peaked in his last years there, then fell. Its rise began in the late 1870s, suggesting gradual politicisation rather than a single editor effect.

Responsible NLP. *Ablate before interpreting:* at L1, ⟨bh⟩ and apostrophes would pass for Imvo's "voice". *Noise is partly culture:* Imvo's spelling (e.g. *Xhosa*, never in Isigidimi) is plausibly editorial; removing it suits questions of topic, not identity, and must be documented (Bender & Friedman, 2018). *Forced-choice models misread time:* a binary classifier assigns every unfamiliar text to a class, so Fig. 2b reflects a temporal shift of unisolated cause, not that 1876 Isigidimi "sounded like" Imvo; such outputs must not become ideological labels (Blodgett et al., 2020).

Limitations. With 38 Isigidimi issues, CIs are wide; matching cannot remove unmeasured confounds; frequency and genre are confounded with source; and the lexicons, built by a non-first-language speaker with precision not yet estimated, impose a modern politics/religion split.

Contribution Statement

This was an individual project; the author list is unchanged from the preliminary statement. Sibonelo Mlambo is responsible for all components: reading the background literature, data exploration, experiment design (hypotheses, controls, splits and metrics), implementation of the pipeline and models, analysis, and writing of this abstract, the ethics checklist and the code documentation. Compared with the preliminary statement, (i) the pretrained multilingual model was replaced by fastText embeddings trained on the corpus, because pretrained weights could not be downloaded in the compute environment; (ii) topic modelling was replaced by feature-family ablations and a validated thematic lexicon, which answered the question more directly; (iii) non-content matching, noise injection, a temporal hold-out and an out-of-sample probe of earlier Isigidimi years were added; and (iv) after review feedback, a year-controlled lexicon comparison, matched-set confidence intervals with per-feature balance diagnostics, a lexicon concordance audit and a fragment/name sensitivity model were added as exploratory follow-up analyses, and several claims were narrowed. AI assistance was used as declared below.

Faculty of Science AI Ethics Statement

I declare that generative AI was used in this project, as permitted by the course brief. Claude (Anthropic), used through an agentic coding environment, assisted with: designing the experimental controls, writing and debugging the Python code in the submission, running the experiments, producing the figures, and drafting the text of this extended abstract, the README and the NeurIPS checklist answers. Every claim and number in the abstract was checked against the experiment outputs (`outputs/results.csv`, `outputs/stats.json`), and historical facts were checked against the cited sources. The isiXhosa glosses are limited to words whose meaning I could verify, and the thematic lexicons were checked against corpus concordances. I have read and understood all code and text in this submission and take full responsibility for it, including any errors. No AI system was used to generate data or annotations for the experiments.

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NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim that (i) raw-text source classification is inflated by non-content shortcuts (spelling and transcription proxies alone reach balanced accuracy 0.85), (ii) a source signal survives all controls (about 0.8), which the paper explicitly does not attribute exclusively to politics, and (iii) a separate lexicon analysis shows more political vocabulary in Imvo, including within every common year. Each claim is tied to Table 1, a figure or a supplementary table. The scope is two newspapers from November 1884 to December 1888, plus earlier Isigidimi issues used only as a probe.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Limitations paragraph, Methods and Supplement note the small number of Isigidimi issues (38; wide confidence intervals), the missing 1886 Isigidimi volume, the hand-built lexicons (drafted by a non-first-language speaker, not reviewed by a first-language speaker, without estimated precision, with several patterns that also carry non-political or non-religious senses, and imposing a modern politics/religion split), the confounding of publication frequency and genre with source, the loose separation of style and topic by word frequency, matching that reduces only measured imbalance, transcription features that are proxies rather than measured OCR error rates, and the absence of pretrained encoders. The Discussion explains why the out-of-sample classifier probe cannot be read as an editor effect, and the follow-up analyses added after inspecting results are labelled exploratory.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

   Justification: The paper is empirical and contains no theoretical results.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The Methods section and Supplement specify the segmentation (200-token windows), the cleaning ladder L0–L4, the issue-grouped 70/15/15 split stratified by paper and year, the near-duplicate filter, the model families and hyper-parameter grids (Supplement S1), the matching procedure and the noise-injection procedure. Random seeds are fixed in the code; fastText is trained single-threaded with a fixed `PYTHONHASHSEED`, so reruns with the pinned package versions are deterministic.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The code, a `requirements.txt` with pinned versions, a `README.txt`, a one-command script (`reproduction.sh`) and a Colab notebook are publicly available at `https://github.com/SiboneloMlambo/NLP_COMS7066A` (folder `Project2_isiXhosa`). The newspaper text itself is not redistributed there (see the licence answer below); the README explains how to obtain and place it.

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: The splits, grids and selection criterion (validation macro-F1, test set used once) are described in Methods; the selected hyper-parameters for every run are written to `outputs/results.csv`.

7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Test scores carry 95% confidence intervals from a cluster bootstrap that resamples whole issues (1,000 resamples); for the matched subset, the matched segments are held fixed and issues are resampled within each newspaper. The standard deviation over three independent issue splits is also reported. Lexicon differences use issue-level bootstrap intervals and permutation tests (within year for the year-controlled comparison).

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments run on CPU only (2 virtual cores, 8 GB RAM, no GPU). The full pipeline (three seeds, all models including single-threaded fastText, plus the grid-edge check) takes about 75 minutes of wall-clock time with `reproduction.sh`, which runs the seeds in two parallel processes; `--fast` skips fastText.

9. **Code Of Ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work uses publicly archived historical newspapers (1884–1888), involves no living human subjects, releases no generative model, and discusses the risks of attributing ideology to historical African authors from noisy text.

10. **Broader Impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Positive: the work gives digital humanities researchers a concrete control protocol before drawing cultural conclusions from OCR-derived isiXhosa archives. Negative: source and stance classifiers trained on such data can turn digitisation artefacts and modern categories into confident but wrong claims about historical Black South African writers and their politics, and can misattribute anonymous or pseudonymous writing; the Discussion addresses this.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: No new dataset or high-risk model is released; the classifiers are small linear models trained for analysis only.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The texts come from the isiXhosa Intellectual Traditions Digital Archive (Department of Sociology, University of Cape Town; Schoots et al., 2023). The archive's item pages state that the Word documents are "downloadable and reusable" without naming a licence, while UCT's eMandulo/Five Hundred Year Archive mirror of the converted text files marks them CC BY-NC-ND 4.0. We therefore use them only for non-commercial coursework, credit the source, and do not redistribute the text or derived corpora in our public repository. High-resolution scans belong to the National Library of South Africa and were not used. Software (scikit-learn, gensim, NumPy, pandas, SciPy, Matplotlib) is open source and listed with versions in `requirements.txt`.

13. **New Assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The only new assets are the analysis code and two small thematic lexicons (in `src/features.py`), documented in the README and in code comments with the gloss of every stem.

14. **Crowdsourcing and Research with Human Subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: No crowdsourcing or human-subject research was performed.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification: No human participants were involved.

Supplementary Material

S1. Implementation details. Segments are 200 whitespace tokens (windows with fewer than 60 raw or 30 L4 tokens are dropped). Grids: word LR $C \in \{0.3, 1, 3, 10, 30\}$, n-grams $\{(1,1),(1,2)\}$, min. df $\{2,5\}$; char LR $C \in \{1,3,10,30\}$, $n \in \{(2,4),(2,5)\}$; Complement NB $\alpha \in \{0.01, 0.03, 0.1, 0.3, 1\}$, min. df $\{2,5\}$; fastText (skip-gram, window 5, subwords 3–5, 10 epochs, one worker thread, `PYTHONHASHSEED=0`, 200k hash buckets) dim $\{50,100\}$ with LR $C \in \{0.01, 0.1, 1\}$; non-content LR $C \in \{10^{-3}, \ldots, 10^{2}\}$; histogram gradient boosting learning rate $\{0.05, 0.1\}$, depth $\{2,3,5\}$. All logistic regressions are class-balanced. Selected values for every run are in `outputs/results.csv`. Software: scikit-learn (Pedregosa et al., 2011), gensim (Řehůřek & Sojka, 2010). Compute: CPU only (2 cores, 8 GB RAM); about 65 minutes wall-clock with the two README commands run in parallel.

S2. Lexicons (patterns matched inside lower-cased tokens; old orthography; 14 political patterns for 10 concepts, 12 religious patterns for 11 concepts). *Political:* palamente (parliament), voti/vota (vote), nyulo/nyulw/nyula (elect, election; excluding *-qanyul-* "cut"), rulumen (government), mteto/miteto (law), lungelo (right), rafu (tax), mantyi (magistrate), tyala (court case, charge), koloni (the Colony). *Religious:* tixo (God), yesu (whole word), kristu (Christ), tandaz (pray), zibalo (scripture), vangeli (gospel), ngcwele (holy), sono (sin, word-final), mfundisi/bafundisi (minister/s), cawa (church, Sunday), nkonzo (service). Patterns marked † in Table 10 also carry other common senses (e.g. *nyula* "choose" goods, *tyala* "debt", *tandaza* petitioning, *cawa* "Sunday", *mfundisi* "teacher"); the strict comparison drops them. A blank annotation sheet (15 random matches per pattern per newspaper) is provided so that precision can be estimated by a first-language reader; it has not yet been completed. The patterns were drafted from general knowledge of isiXhosa before any per-paper comparison; the concordance audit (Table 10) removed or anchored patterns with false matches (Table 11), and none was added or removed because of the direction of a difference. *bhunga* (council) was dropped because Isigidimi never writes ⟨bh⟩, so it could only match Imvo's spelling. The glosses are the author's and have not been reviewed by a first-language speaker.

S3. Synthetic noise. The noisier paper was identified by its rate of vowel-less plus mixed letter–digit tokens (Imvo 3.8% vs. Isigidimi 1.4% of tokens; rare-token rate was not used because it is confounded with each paper's share of the training vocabulary). Confusions were injected into Isigidimi test segments at 0–40% of tokens. Fig. 4 reports per-segment flips: the share of segments whose prediction changes from Isigidimi to Imvo relative to the uncorrupted text.

S4. Exploratory follow-ups. The fragment/name sensitivity model, the issue-level intervals and balance diagnostics for the matched set, the year-controlled and strict lexicon comparisons and the lexicon audit were added after inspecting the main results and an external review. They were not part of the original design and should be read as exploratory.

Table 2: Cleaning ladder, test macro-F1 / balanced accuracy, mean ± s.d. over three issue-grouped splits (seeds 13, 21, 42).

Level	Model	Macro-F1	Bal. acc.
L0	char	0.937 ± 0.010	0.929 ± 0.007
L0	fastText	0.829 ± 0.028	0.866 ± 0.014
L0	CNB	0.812 ± 0.029	0.875 ± 0.020
L0	word	0.902 ± 0.008	0.887 ± 0.006
L1	char	0.927 ± 0.012	0.917 ± 0.017
L1	fastText	0.807 ± 0.015	0.830 ± 0.007
L1	CNB	0.857 ± 0.021	0.861 ± 0.010
L1	word	0.891 ± 0.009	0.884 ± 0.011
L2	char	0.916 ± 0.017	0.905 ± 0.020
L2	fastText	0.798 ± 0.014	0.817 ± 0.017
L2	CNB	0.852 ± 0.016	0.854 ± 0.005
L2	word	0.886 ± 0.013	0.872 ± 0.020
L3	char	0.906 ± 0.025	0.896 ± 0.029
L3	fastText	0.758 ± 0.025	0.783 ± 0.022
L3	CNB	0.819 ± 0.013	0.822 ± 0.010
L3	word	0.859 ± 0.016	0.846 ± 0.020
L4	char	0.816 ± 0.007	0.818 ± 0.015
L4	fastText	0.737 ± 0.018	0.768 ± 0.005
L4	CNB	0.767 ± 0.009	0.801 ± 0.017
L4	word	0.839 ± 0.019	0.829 ± 0.026

Table 3: Robustness (seed 13). *No dedup*: near-duplicate filter switched off. *Temporal*: train/validate on Nov 1884–1887, test on 1888. Reference: the main random split.

Run	Level	Model	Macro-F1	Ref. macro-F1
No dedup	L0	word	0.902	0.899
No dedup	L0	char	0.928	0.926
No dedup	L1	word	0.890	0.883
No dedup	L1	char	0.920	0.916
Temporal	L1	word	0.864	0.883
Temporal	L1	char	0.896	0.916
Temporal	L4	word	0.787	0.840
Temporal	L4	char	0.745	0.811

Table 4: Grid-edge check (validation macro-F1, seed 13). Several selected C values sat at the top of the grid; larger values do not improve validation scores.

Level	Model	C=10	30	100	300
L1	char	0.944	0.948	0.946	0.946
L1	word	0.885	0.888	0.890	0.886
L4	char	0.824	0.813	0.798	0.795
L4	word	0.836	0.824	0.817	0.815

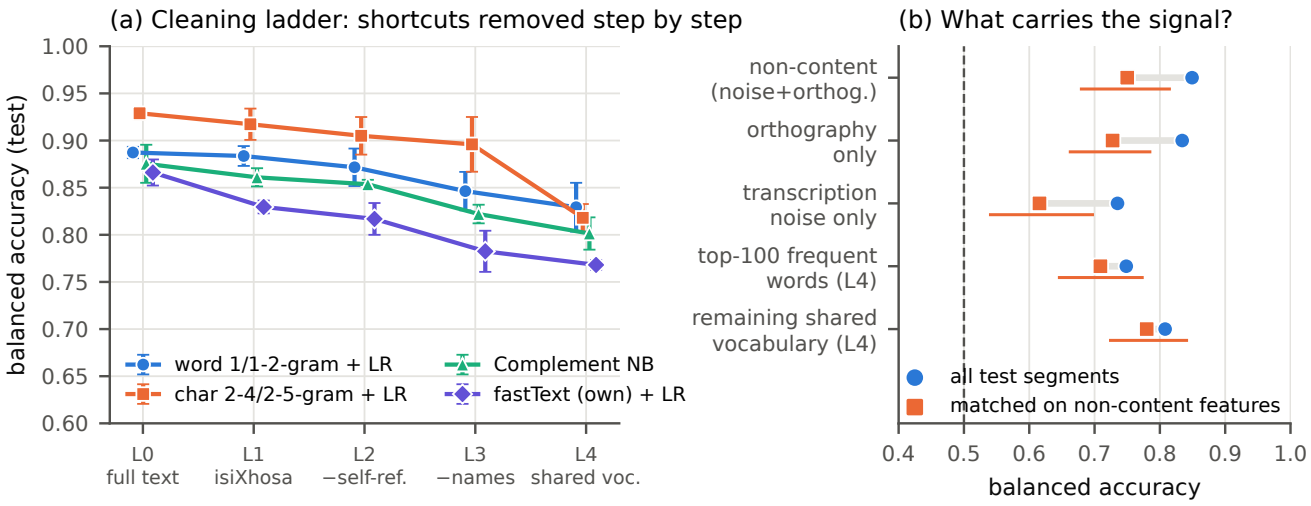


Figure 3: Extended version of Fig. 1. **(b)** Single feature families (seed 13) on all test segments (blue) and on the 134 pairs matched for the 19 non-content features (orange): matching costs the remaining-vocabulary model 0.03 but the non-content models 0.10–0.12. Thin lines: 95% CIs for matched accuracy (resampling issues within each newspaper).

Table 5: Editorial eras. Lexicon rates per 1,000 tokens (issue-level means, bootstrap 95% CI over issues; all issues). Probe: mean P(Imvo) from the L4 content model and share of segments classified Imvo (post-1884 rows: held-out test issues only).

Group	Issues (lex.)	Political/1k	Religious/1k	Political share	Mean P(Imvo)	% segs. Imvo
Isigidimi 1876-80 (pre-Jabavu)	51	8.93 [8.12, 9.80]	8.23 [7.08, 9.40]	0.54 [0.49, 0.58]	0.55	58
Isigidimi 1881-Oct 1884 (Jabavu)	56	10.04 [9.00, 11.12]	4.76 [4.22, 5.34]	0.67 [0.63, 0.70]	0.60	66
Isigidimi Nov 1884-88 (post-Jabavu)	38	7.38 [6.22, 8.73]	5.14 [4.32, 5.98]	0.59 [0.54, 0.63]	0.34	26
Imvo Nov 1884-88	211	13.97 [13.24, 14.70]	3.23 [3.01, 3.47]	0.80 [0.78, 0.81]	0.78	88

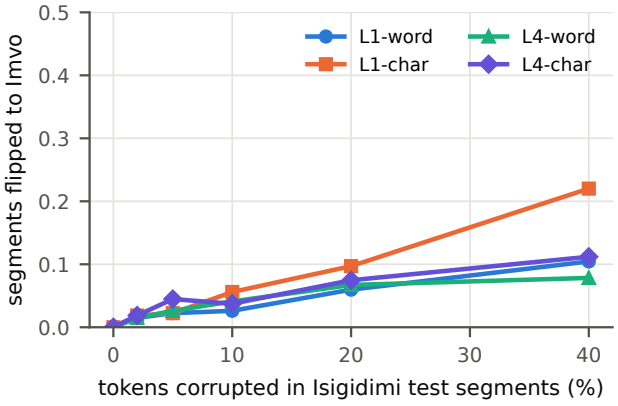


Figure 4: Share of Isigidimi test segments whose prediction flips to Imvo as synthetic OCR confusions are injected. At 2–5% corruption 2–5% of segments flip; only heavy corruption moves the raw character model substantially.

Table 6: Twenty largest-weight features of the L4 content model, in order (seed 13). Glosses for selected items are given in the Results.

Favour Imvo	Favour Isigidimi
njengokuba, abamnyama, mayelana, base, ase, uku, sprigg, egqitileyo, kukuba, ba, zetu, epepeni, suka, omnyama, ixabiso, oka, okokuba, lokuba, amawetu, kwelilizwe	*imbali, ed, ngoko, ngosuku, malunga, uzele, suke, ezinje, londawo, inani, um, kanjalo, mbali, evekini, umlambo, ngolwe, ingu, ngxoxo, kwase, ndawo*

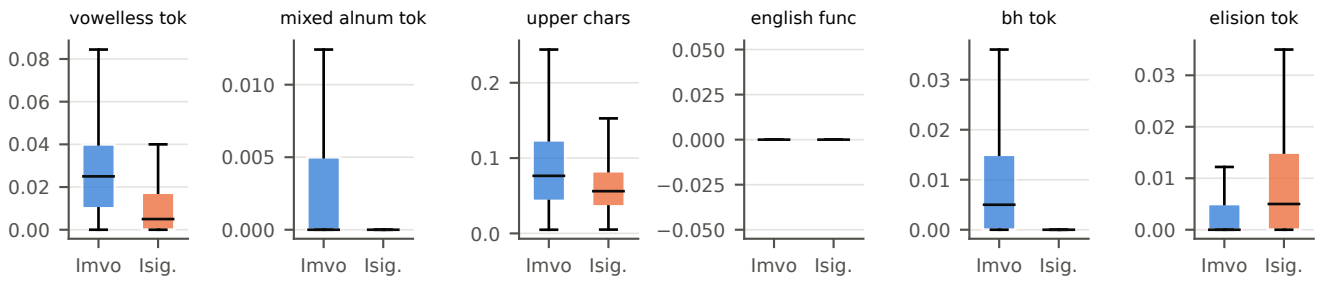


Figure 5: Selected non-content features per segment (training split; boxes: interquartile range, whiskers: 1.5 IQR). Imvo has more vowel-less and mixed letter–digit tokens, uppercase characters and English function words; the ⟨bh⟩ digraph is almost exclusive to Imvo and apostrophe elision is more frequent in Isigidimi.

Table 7: Matched evaluation (seed 13): balanced accuracy on all test segments and on the matched pairs, with a 95% CI from resampling whole issues within each newspaper while holding the matched segments fixed (1,000 resamples).

Model	BA (all)	BA (matched)	95% CI
L1 words	0.880	0.854	[0.81, 0.91]
L1 char	0.906	0.862	[0.81, 0.92]
L2 words	0.857	0.832	[0.78, 0.90]
L2 char	0.897	0.851	[0.80, 0.91]
L3 words	0.847	0.802	[0.74, 0.86]
L3 char	0.880	0.832	[0.78, 0.89]
L4 words	0.837	0.810	[0.75, 0.87]
L4 char	0.814	0.806	[0.74, 0.87]
Transcription (LR)	0.687	0.578	[0.51, 0.65]
Transcription (GB)	0.735	0.616	[0.54, 0.70]
Orthography (LR)	0.831	0.735	[0.68, 0.79]
Orthography (GB)	0.834	0.728	[0.66, 0.78]
All non-content (LR)	0.858	0.750	[0.69, 0.80]
All non-content (GB)	0.849	0.750	[0.68, 0.81]
L4 remaining vocabulary	0.808	0.780	[0.72, 0.84]
L4 top-100 frequent words	0.749	0.709	[0.65, 0.77]
L4 remaining, minus fragments/names	0.779	0.746	[0.68, 0.82]

Table 8: Per-feature balance before and after matching: difference in means (Imvo − Isigidimi) in training-standard-deviation units. 134 pairs from 35 issues; mean |SMD| 0.29 → 0.08; 1 feature(s) still exceed 0.25 after matching.

Feature	Before	After
mean_tok_len	+0.05	+0.13
sd_tok_len	+0.50	+0.18
short_tok	+0.30	+0.07
long_tok	+0.20	+0.03
upper_chars	+0.10	-0.11
punct_chars	-0.38	-0.10
digit_chars	+0.08	-0.04
nonascii_chars	+0.27	+0.04
vowelless_tok	+0.28	+0.08
mixed_alnum_tok	+0.08	-0.01
rare_tok	-0.40	-0.08
type_token_ratio	-0.16	-0.02
english_func	+0.27	-0.02
cap_initial_tok	+0.24	+0.03
split_fragment	-0.01	+0.00
bh_tok	+0.82	+0.29
xh_tok	+0.54	+0.04
elision_tok	-0.71	-0.22
hyphen_tok	-0.10	-0.09

Table 9: Year-controlled lexicon comparison: years in which both papers appear (Isigidimi 1886 is missing; 1884 = Nov–Dec). Issue-level means; CI of the difference from bootstrapping issues within the year. *Strict*: patterns with other common senses removed (nyulo/nyulw/nyula, tyala, tandaz, cawa, mfundisi/bafundisi). Exploratory follow-up analysis. Political / 1k: mean within-year difference +5.91, within-year permutation $p = 0.0002$; Religious / 1k: mean within-year difference -2.25, within-year permutation $p = 0.0002$; Political share: mean within-year difference +0.23, within-year permutation $p = 0.0002$; Political / 1k, strict: mean within-year difference +4.78, within-year permutation $p = 0.0002$; Religious / 1k, strict: mean within-year difference -1.99, within-year permutation $p = 0.0002$; Political share, strict: mean within-year difference +0.26, within-year permutation $p = 0.0002$.

Measure	Year	Imvo	Isigidimi	Difference	95% CI	Issues (I/Is)
Political / 1k	1884	10.85	6.55	+4.30	[1.41, 6.97]	9/2
Political / 1k	1885	11.88	5.74	+6.14	[4.13, 7.90]	49/12
Political / 1k	1887	15.84	8.75	+7.10	[4.11, 9.70]	49/12
Political / 1k	1888	13.88	7.78	+6.10	[2.83, 9.08]	52/12
Religious / 1k	1884	1.90	5.94	-4.04	[-5.36, -2.73]	9/2
Religious / 1k	1885	3.32	5.51	-2.19	[-3.46, -0.86]	49/12
Religious / 1k	1887	3.02	3.59	-0.57	[-1.73, 0.53]	49/12
Religious / 1k	1888	3.97	6.19	-2.21	[-4.06, -0.43]	52/12
Political share	1884	0.83	0.52	+0.31	[0.26, 0.36]	9/2
Political share	1885	0.77	0.51	+0.26	[0.17, 0.35]	49/12
Political share	1887	0.83	0.71	+0.13	[0.06, 0.19]	49/12
Political share	1888	0.76	0.56	+0.20	[0.11, 0.29]	52/12
Political / 1k, strict	1884	8.42	5.44	+2.97	[0.76, 5.04]	9/2
Political / 1k, strict	1885	9.10	4.19	+4.91	[3.55, 6.29]	49/12
Political / 1k, strict	1887	13.47	7.47	+6.00	[2.95, 8.56]	49/12
Political / 1k, strict	1888	11.00	5.76	+5.24	[2.35, 7.87]	52/12
Religious / 1k, strict	1884	0.27	3.45	-3.18	[-3.51, -2.84]	9/2
Religious / 1k, strict	1885	0.73	2.31	-1.58	[-2.16, -0.91]	49/12
Religious / 1k, strict	1887	0.35	1.29	-0.94	[-1.42, -0.53]	49/12
Religious / 1k, strict	1888	1.01	3.27	-2.26	[-3.54, -1.08]	52/12
Political share, strict	1884	0.96	0.61	+0.35	[0.30, 0.40]	9/2
Political share, strict	1885	0.92	0.64	+0.28	[0.18, 0.39]	49/12
Political share, strict	1887	0.97	0.83	+0.14	[0.09, 0.19]	49/12
Political share, strict	1888	0.90	0.61	+0.29	[0.22, 0.36]	52/12

Table 10: Lexicon audit (main window). For each pattern: matches in each paper, the five most frequent matched word types, and one randomly drawn concordance line (lower-cased, punctuation removed). Patterns were drafted from general knowledge of isiXhosa before any per-paper comparison; the concordance check removed false matches (Table 11) and no pattern was added or removed because of the direction of a difference. †: pattern with other common senses (excluded in the strict comparison). The audit documents matches but is not semantic validation: precision has not been estimated, and a blank annotation sheet (`lexicon_annotation_sheet.csv`) is provided for a first-language reader. Glosses are the author's.

Cat.	Pattern	Imvo	Isig.	Top matched types	Example context
pol.	`koloni`	1117	159	*koloni (1108); ikoloni (89); yasekoloni (22); mkoloni (14); umkoloni (9)*	*lase bechuanaland liya kuhlonyelwa kule koloni nokuba u montsiwa seleteta ntonina*
pol.	`lungelo`	1139	212	*amalungelo (346); ilungelo (294); lungelo (94); nelungelo (56); namalungelo (51)*	*ngamatanda mfazwe encediswa zizihlobo mpi namalungelo abo ilungelo la mampondo ubelenze*
pol.	`mantyi`	1847	155	*mantyi (605); imantyi (487); yimantyi (128); umantyi (126); yemantyi (94)*	*emva kokutabata inteto yetu engokuba imantyi iyakuhlala esihlalweni ngomso edikeni ukuva*
pol.	`miteto`	487	65	*imiteto (245); miteto (96); lemiteto (34); nemiteto (28); ngemiteto (22)*	*imimiselo nemi gqaliselo nokuba yi miteto nokuba zinxalenye zayo ecaseneyo mhlaimbi*
pol.	`mteto`	2206	278	*umteto (727); lomteto (490); mteto (368); ngomteto (173); ngumteto (99)*	*nokupendula imicimbi yombuso banayo imfanelo ngomteto wokunyula owebala labo eze kulenkundla*
pol.	`nyula`†	324	41	*lokunyula (65); ukunyula (39); onganyula (24); yokunyula (16); ngokunyula (13)*	*bacelwe ukuba banganqopisani namntu ungoyena bayakumnyula ade onke amadoda ayakuma abe*
pol.	`nyulo`†	232	32	*unyulo (73); wonyulo (28); lonyulo (18); yonyulo (18); nyulo (17)*	*eziyakuze zibakanyele i feltkornet ngomhla wonyulo kuba zoti zakuba zibuze igama*
pol.	`nyulw`†	548	68	*kunyulwe (89); enyulwe (64); kunyulwa (45); ukunyulwa (32); anyulwe (30)*	*u faku col griffith yena anyulwe eko lomadinga anjalo esenziwa ngawayesilwa*
pol.	`palamente`	1842	114	*palamente (1857); ipalamente (46); palamenteni (31); epalamente (15); bapalamente (1)*	*kokuba ade abe uyalelwe yi palamente u mr j umnumzetu unga*
pol.	`rafu`	874	84	*irafu (400); rafu (347); lerafu (39); yerafu (35); lorafu (23)*	*read lokuba bazihlaule ngexesha i rafu zabo kuba kungenakubako lulaulo ngapandle*
pol.	`rulumen`	3124	318	*rulumente (1996); rulumeni (1410); rulumenteni (7); urulumente (5); urulumeni (3)*	*gaso ' akasoia aye ku rulumeni ' makabe u qipu yinkosi*
pol.	`tyala`†	2349	300	*ityala (476); amatyala (449); tyala (247); yamatyala (138); ngetyala (66)*	*laya ixelwe ekuqaleni ngokubheki selele kumatyala okwapula iziqendwana ezi zinga a*
pol.	`vota`	202	32	*lokuvota (54); kuvota (19); ukuvota (18); vota (15); yokuvota (14)*	*abantsundu ngabenga nikwanga nelungelo loku vota ayitandi inxenye yakubona sivota ke*
pol.	`voti`	1051	175	*ivoti (299); voti (207); evotini (140); abavoti (99); ngevoti (56)*	*lonto kuti amawetu anevoti makangake anikele indlebe zawo konomgogwana*
rel.	`^yesu$`	30	48	*yesu (78)*	*ibuhlungu kuba ndilele apa no yesu inkosi yam ute xa bekusondela*
rel.	`bafundisi`†	952	281	*abafundisi (436); yabafundisi (137); ngabafundisi (97); bafundisi (81); nabafundisi (77)*	*ke ndiveleni obubunzima zinkosi zam bafundisi base wesileni ndiyanitandaza ndiyazitoba ndisisicaka*
rel.	`cawa`†	674	92	*cawa (478); ingcawa (156); ngcawa (64); yencawa (12); ngecawa (7)*	*ibokwe inani lotnkosi ngomhla we cawa dec wawumayelana senza ityalike enkulu*
rel.	`kristu`	80	151	*kristu (206); makristu (8); kristukazi (8); mkristu (3); amakristu (2)*	*siyibona ukwapulwa kwayo lemimiselo ka kristu kwi eamente*
rel.	`mfundisi`†	932	231	*umfundisi (488); mfundisi (162); ngumfundisi (108); nomfundisi (55); yomfundisi (48)*	*nokuba ngabapina ngapandle kwamatanda into egumfundisi omkulu ngezimini endaba besingati simi*
rel.	`ngcwele`	75	42	*oyingcwele (29); ongcwele (17); ezingcwele (16); ngcwele (8); ngcweleshe (6)*	*ngokufunda amaculo la ati yerusalem engcwele eyase zulwini ndilundwendwe ndingumhambi ezulwini*
rel.	`nkonzo`	107	27	*inkonzo (85); nkonzo (8); lenkonzo (6); kwinkonzo (5); kwenkonzo (4)*	*zatakazelwa kunene kulandawo yaqalwa kuyo lenkonzo ka mahommet babesiti namhla olu*
rel.	`sono$`	89	25	*isono (42); sisono (20); sono (12); ngesono (8); nesono (8)*	*nazintsizi kwa nazifo asiko kwa nesono akuko maxala nangqumbo akuko sipelo*
rel.	`tandaz`†	678	169	*umtandazo (146); imitandazo (76); ngomtandazo (73); abatandazi (34); lomtandazo (22)*	*october ozayo intlanganiso ivalwe ngo mtandazo j a v port elisabeth*
rel.	`tixo`	394	291	*tixo (678); otixo (1); botixo (1); bezitixo (1); likatixo (1)*	*eyirola imali ngokwaneleyo wanga u tixo angasikelela kumzamo wenu lentlanganiso yangena*
rel.	`vangeli`	76	34	*vangeli (39); umvangeli (21); ivangeli (14); mvangeli (12); abavangeli (6)*	*ugama lingezantsi nfuna u m vangeli ne titshala ezimbini ezine certificates*
rel.	`zibalo`	14	25	*izibalo (17); zibalo (11); yezibalo (4); bezibalo (1); zezibalo (1)*	*ilahlekise izilumko ezilinani malunga nentyilo yezibalo commentators yiyo le ke kanye*

Table 11: Candidate stems rejected or moved after the concordance check.

Stem	Reason
kolwa	mostly 'kolwandle' (sea); 'kolwa' (believe) is not specifically religious
bunga	mostly unrelated words ('bungane', '-bungatengiswa')
lawul	mostly 'hlawula' (pay, fine)
yesu	unanchored form matches 'yesuka' (start); kept only as a whole word
nyulwa (religious list)	means 'be elected'; belongs to the political list
bhunga	orthography-specific: only Imvo writes ¡bh¿

Table 12: Tokens removed from the remaining shared vocabulary in the sensitivity model: short tokens ($\leq$3 letters), tokens that precede a masked name in $\geq$50% of training occurrences (separated prefixes), and tokens capitalised in $\geq$80% of raw occurrences (residual names). Examples are the most frequent in each group.

Reason	n	Examples
capitalised	45	*sprigg, yizani, iyeza, umbona, uhlala, pamla, mzimba, mhleli, bonisani, sakuba, rubusana, amazimba*
prefix-before-name	129	*wase, yase, lase, nase, ngolwesi, base, lika, yama, ngama, nama, sase, zase*
short	222	*ke, no, ka, ngu, to, ne, ngo, xa, ku, kwa, nge, apa*

APPENDIX B:

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☑
2. Sourcing Related Work (summarising, identifying sources) ☑
3. Methods and Experiment Design (experiment setup, model tuning) ☑
4. Data Analysis (presentation, coding, interpretation) ☑
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☑
7. Presentation (rendering graphics, formatting) ☑
8. Editing (grammar, readability) ☑
9. Writing (text generation, document structuring) ☑
10. Citation Formatting (structuring, organising) ☑

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**
Claude (Anthropic)	Items 1-4, 6-10 above; see AI Ethics Statement

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 536187

Candidate signature: Sibonelo_Mlambo

Date: 2026-10-29